

SmartSpend

AI-Powered Financial Fraud Detection & Risk Analytics Platform

TSM-TECHNOVA 2026 Presentation Submission

Candidate: Sidhesh | Track: Fintech, Cybersecurity & AI

Status: Working Prototype Ready

The Challenge

The Real-Time Digital Payment Fraud Crisis

- **Sub-Second Settlement Speed:** UPI and IMPS clear funds in <200ms. Illicit transactions leave the bank perimeter before traditional audits run.
- **Organized Money Mule Rings:** Scammers rapidly bounce stolen funds through 4 to 6 intermediary accounts in circles to evade tracking.
- **Legacy Brittle Rules:** Static thresholds generate thousands of false alarms, frustrating genuine customers.
- **Black-Box AI Limitations:** Traditional machine learning outputs opaque numbers without explainability, causing investigation fatigue.

SmartSpend: Dual-Stage Pre-Settlement Interception

SmartSpend operates as an intelligent pre-settlement guard positioned between payment gateways and bank ledgers:

- **Stage 1: Machine Learning Classifier:** Evaluates 17 behavioral signals to score risk in <50ms.
- **Stage 2: Fraud Ring Network Tracker:** Traces multi-hop circular laundering loops across accounts.
- **Dynamic Policy Automation:**
 - **0 – 30 (Low Risk):** Instant automated settlement.
 - **31 – 70 (Medium Risk):** Step-up OTP / Biometric authentication.
 - **71 – 100 (High Risk):** Pre-settlement hold and fraud analyst triage.

Validated ML Performance on 12,000 Holdout Tests

Accuracy	Fraud Recall	Precision	F1-Score	ROC-AUC
98.74%	90.91%	65.40%	0.7607	0.9973

Holdout Confusion Matrix (Unseen Test Data):

- **11,609 True Negatives:** Genuine payments cleared without friction.
- **127 False Positives:** Legitimate full-balance payments flagged for safety review (reflects real banking behavior).
- **24 False Negatives:** Subtle micro-probing fraud attempts.
- **240 True Positives:** Confirmed high-risk fraud attacks intercepted before settlement.

Fraud Ring Network Tracker (Multi-Hop Interceptor)

- **The Money Mule Challenge:** Individual transfers look innocent in isolation. Criminals move money in circles ($A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$) to hide origins.
- **In-Memory Traversal:** Discovers closed circular loops and smurfing funnels in <15ms without slow database re-indexing.
- **Interactive Visual Map:** Frontend SVG network graph visualizes account connections, money flows in INR (₹), and highlighted criminal loops.
- **One-Click Ring Freeze:** Executive managers can freeze all accounts in a detected ring with a single click.

Role-Based Banking Governance Workflows

- **Sidhesh (Analyst) — Frontline Investigator:**

- Reviews flagged transaction alerts and inspects sender/receiver ledger movement.
- Can **Approve** verified payments or **Request OTP / MFA**.
- ■ Cannot freeze accounts (strict regulatory safeguard).

- **Sidhesh (Manager) — Executive Risk Authority:**

- Full supervisory oversight across all accounts.
- ■ **Authorized to permanently Block & Freeze accounts and entire fraud rings.**
- Exports compliance audit logs for central bank (RBI) reporting.

Conclusion & Impact

Economic Impact, UN SDGs & Working Prototype

- **Economic Value:** Stops capital loss before settlement; reduces manual investigation overhead by 80%.
- **Social Trust:** Protects vulnerable citizens from cyber scams; fortifies digital public infrastructure.
- **UN SDGs:** Advances SDG 8 (Decent Work), SDG 9 (Infrastructure), and SDG 16 (Anti-Illicit Financial Flows).

Working Prototype: Live and accessible at <http://localhost:3000>

Thank you for your time and consideration!